

Spatio–Temporal Predictive Modelling of PM_{2.5} in Karachi from Multi-Source Satellite, Meteorological and Ground Observations (2019–2023)

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Abstract

Karachi’s ambient PM_{2.5} concentrations average approximately 12.3 times the World Health Organization 2021 annual guideline of $5\text{ }\mu\text{g m}^{-3}$ on the 2019–2023 dataset analysed in this study, placing it among the most polluted megacities in South Asia. Systematic, data-driven characterisation of the city’s pollution sources and temporal structure remains limited in the peer-reviewed literature. We address this gap by constructing a five-year daily spatio-temporal dataset from eight ground monitoring stations fused with multi-source satellite retrievals (Sentinel-5P trace gases, MODIS MAIAC aerosol optical depth, ERA5 meteorological reanalysis, VIIRS nighttime lights) and a strict ground-truth provenance chain that combines OpenAQ per-station observations, a US Consulate reference sensor, and MERRA-2 citywide surface PM_{2.5}. The final modelling set contains 14 400 station-day observations across 13 feature columns. We trained five traditional machine-learning models and a unidirectional causal LSTM with self-attention on a 2019–2022 training set, evaluated on a held-out 2023 test set that the models never saw. Random Forest achieved the best predictive performance (RMSE = $16.30\text{ }\mu\text{g m}^{-3}$, MAE = $10.63\text{ }\mu\text{g m}^{-3}$, $R^2 = 0.612$). The LSTM, despite approximately 480 000 trainable parameters and a 14-day lookahead, produced a negative R^2 of -0.104 on the test set, a result we interpret as evidence of a fundamental data-density mismatch between complex sequence architectures and eight monitoring stations. SHAP analysis of the Random Forest identified `pm25_lag1` and `Optical_Depth_055` as the dominant predictors, indicating that pollution persistence and aerosol loading explain most of the predictable variance. Moran’s I on the RF residuals is -0.19 ($p = 0.28$, not significant), confirming that no detectable spatial autocorrelation remains after the RF absorbs the satellite and meteorological features — a positive finding about the model’s explanatory power rather than a defect. We release the full preprocessing pipeline, processed dataset, trained model weights, and an interactive 3D digital twin as an open-source repository intended to be reusable for other data-sparse South Asian cities.

1 Introduction

1.1 The Karachi air quality context

Air pollution remains one of the leading contributors to the global burden of disease. The Global Burden of Disease 2019 attributed approximately 6.7 million deaths to ambient and household air pollution combined [1], and the burden falls disproportionately on South and East Asian cities where industrial growth has outpaced regulatory enforcement. In South Asia, the Indo-Gangetic Plain cities — Delhi, Lahore, Dhaka — receive the most research attention. Karachi, Pakistan’s commercial capital and largest city, is discussed far less in the peer-reviewed literature, yet its exposure levels are comparable to the most polluted cities in the region.

Karachi is home to an estimated 16–20 million people. It hosts Pakistan’s largest port and industrial base, including steel mills, petrochemical plants, textile factories, and power-generation units concentrated in its northern and western industrial zones. The city’s meteorology compounds the problem: surrounded on the south by the Arabian Sea but hemmed inland by flat terrain, Karachi experiences persistent atmospheric inversions in winter months that trap ground-level

emissions. The annual average $\text{PM}_{2.5}$ concentration on the eight-station dataset used in this study is approximately $61 \mu\text{g m}^{-3}$, compared to the WHO 2021 annual guideline of $5 \mu\text{g m}^{-3}$ — a factor of approximately 12.3.

1.2 Gaps in existing work

Most published air quality studies for Pakistani cities rely on one of two approaches: receptor modelling (PMF, CMB) on short measurement campaigns at single sites, or single-sensor satellite retrievals calibrated against ground truth from outside the region. Both approaches have well-known limitations. Receptor modelling requires high-resolution chemical speciation data that long-term monitoring networks in Pakistan do not routinely provide. Single-sensor satellite approaches — typically MODIS AOD at 10 km resolution — miss the intra-urban variation that drives unequal exposure across neighbourhoods.

Spatio-temporal machine-learning frameworks that fuse multi-satellite data with ground observations over multi-year periods have become standard in studies of Chinese, Indian and European cities [2, 3, 4]. Pakistan, and specifically Karachi, has no equivalent published study. The closest are national-scale AOD-to-PM regression models for the Indo-Pakistan region [5, 6], which smooth over the spatial heterogeneity of a megacity entirely.

The absence of a rigorous, open, multi-year ML dataset and modelling pipeline for Karachi is the direct gap this work addresses.

1.3 Contributions

We make four specific contributions.

First, we build and publicly release a five-year daily spatio-temporal dataset for Karachi, merging ground $\text{PM}_{2.5}$ measurements from eight stations with satellite retrievals from five distinct sensors and ERA5 meteorological reanalysis. The complete preprocessing pipeline, from raw Google Earth Engine (GEE) extraction to the final modelling matrix, is available at <https://github.com/SidhartSami/karachi-spatio-temporal-airq>.

Second, we systematically compare five traditional machine-learning models and a unidirectional causal LSTM with self-attention on a strict temporal hold-out, and report honest test-set performance including the LSTM’s failure to beat a tree ensemble. We use SHAP (TreeExplainer) to make the Random Forest’s decision logic interpretable.

Third, we conduct a Moran’s I and LISA spatial-autocorrelation analysis on the RF residuals — the appropriate test for whether genuine spatial structure remains after the model absorbs the satellite and meteorological features — and report the resulting null result honestly.

Fourth, we build an interactive 3D digital twin that translates the predictive model into an exploratory tool, allowing users to inspect the 2023 daily $\text{PM}_{2.5}$ field and contrast it against observed values at the eight monitoring stations.

1.4 Paper structure

Section 2 describes the study area, the ground-truth provenance chain, the satellite data sources, and the acquisition pipeline. Section 3 presents the methodology — feature engineering, model architectures, evaluation protocol, interpretability methods, and the spatial-statistics framework. Section 4 reports descriptive statistics, the model-comparison results, the SHAP analysis, the spatial analysis, and the LSTM’s negative result. Section 5 discusses findings, the limitations imposed by the data-density of the eight-station network, and compares the results with regional literature. Section 6 concludes.

2 Data

2.1 Study area

Karachi occupies approximately $3\,527 \text{ km}^2$ along Pakistan’s southern coastline at approximately 24.9°N , 67.1°E . The terrain is largely flat, punctuated by the Kirthar Range foothills to the north and the mangrove-lined creeks of the Indus delta to the east. The Arabian Sea moderates summer temperatures but contributes humidity that affects aerosol hygroscopic growth.

The city’s industrial geography shapes its pollution distribution in predictable but insufficiently documented ways. SITE Industrial Area in the north-west and Korangi Industrial Area in the south-east are the two densest concentrations of heavy industry in the eight-station network. Beyond formal industry, vehicle density on the Karachi road network is extreme: an estimated 7.5 million registered vehicles with a fleet age profile dominated by 1980s and 1990s technology, generating disproportionate per-vehicle emissions. Open waste burning, common in peri-urban areas, adds a diffuse source difficult to capture in any point-source inventory.

Table 1. Karachi monitoring stations used in this study. Coordinates and zone types are stored in `script/stations.json`, the single source of truth for all pipeline scripts.

Station	Lat (°N)	Lon (°E)	Zone
Gulshan-e-Iqbal	24.9056	67.0822	residential
Saddar	24.8560	67.0100	commercial
SITE_Industrial	24.9400	66.9800	industrial
Korangi_Industrial	24.8200	67.0300	industrial
North_Nazimabad	24.9800	67.1200	residential
Gulistan_Jauhar	24.8900	67.1300	residential
Landhi	24.8100	66.9900	industrial
Federal_B_Area	24.9200	67.0500	mixed

Table 2. Ground-truth provenance breakdown for the 14 400 station-day observations. The MERRA-2 fallback is citywide (the same daily value for all eight stations), which limits the per-station signal the modelling set can express.

pm25_source	Rows	Share
merra2.citywide	12 793	88.8%
openaq_us_consulate	1 328	9.2%
openaq_exact	279	1.9%
Total	14 400	100.0%

Meteorologically, the year breaks into distinct regimes relevant to air quality. The south-west monsoon (June–September) brings wet scavenging and increased wind mixing. Winter months (November–February) bring stable atmospheric conditions and the temperature inversions that trap vehicle and industrial emissions near the surface. Spring and autumn are transitional, with periodic dust intrusions from the Thar Desert and Balochistan adding episodic coarse particle loading.

2.2 Ground monitoring network and provenance chain

The eight monitoring stations used in this study are listed in Table 1. Their locations span Karachi’s main land-use zones: three industrial (SITE_Industrial, Korangi_Industrial, Landhi), three residential (Gulshan-e-Iqbal, North_Nazimabad, Gulistan_Jauhar), one mixed (Federal_B_Area) and one commercial (Saddar).

Ground PM_{2.5} for each station-day is sourced via a strict fallback hierarchy. Every row in `master_dataset.csv` carries a `pm25_source` provenance flag drawn from `{openaq_exact, openaq_us_consulate, merra2_citywide, missing}`. The hierarchy is applied in the order below and a row is excluded from the modelling set only when `pm25_source` resolves to `missing`.

1. **OpenAQ v3 per-station** (`pm25_source='openaq_exact'`). For each station, we query OpenAQ within a 25 km radius and select the closest sensor reporting hourly PM_{2.5}. The per-station reading is preferred when available.
2. **US Consulate Karachi anchor** (`pm25_source='openaq_us_consulate'`). When a per-station reading is unavailable, we use the US Consulate Karachi continuous monitoring reading as a citywide benchmark. It is a single fixed location and is therefore assigned to all eight stations for the same day.
3. **MERRA-2 citywide scalar** (`pm25_source='merra2_citywide'`). When both per-station and Consulate readings are unavailable, we use the MERRA-2 surface PM_{2.5} estimate for the Karachi grid cell, derived from the van Donkelaar 2010 hygroscopicity formula applied to MERRA-2 aerosol mass concentrations (BCSMAS, OCSMAS, SO4SMAS, SSSMAS25, DUSMAS25, NIESMAS). The value is a single daily scalar assigned to all eight stations.

The full 2019–2023 dataset (1 826 days) yields 14 400 station-day observations before modelling filters. The provenance breakdown is shown in Table 2. We acknowledge that the MERRA-2 citywide fallback dominates the dataset, and discuss the implications of this data-density limitation for both the modelling results and the spatial analysis in Section 5.

2.3 Satellite remote sensing data

We extracted five categories of satellite-derived variables via the GEE Python API. For each monitoring station we defined a 3 km circular buffer and computed spatial means of each satellite product within that buffer on a daily basis.

Sentinel-5P (TROPOMI). Daily Level-3 column densities for NO₂, SO₂, CO, HCHO and aerosol index at approximately 3.5 km × 5.5 km resolution. We retain the aerosol index (`aer_ai`) and the trace-gas columns as proxy signals for traffic/industrial combustion, sulphur sources and carbonaceous loading.

MODIS MCD19A2 (MAIAC). Daily 1 km aerosol optical depth at 0.47 µm and 0.55 µm (`Optical_Depth_047`, `Optical_Depth_055`). These are the fine-mode-sensitive bands most correlated with PM_{2.5} in the literature for South Asian urban environments [5]. The 1 km MAIAC product offers substantially better urban spatial resolution than the standard 10 km Dark Target retrieval.

ERA5. Hourly meteorological fields from the ECMWF reanalysis at 0.25° resolution, aggregated to daily: 10-m wind speed and direction, 2-m temperature and dewpoint, relative humidity, surface pressure and boundary-layer height. Boundary-layer height proved informative as a dilution proxy.

VIIRS Nighttime Lights. Monthly composites of nighttime radiance as a socioeconomic and activity-intensity proxy, assigned to each station’s spatial cell.

Sentinel-2 MSI. Bi-monthly composites for NDVI and NDBI (Normalized Difference Built-up Index) at 10 m resolution, spatially aggregated to a 3 km buffer around each monitoring station.

2.4 Preprocessing pipeline

Raw data quality varies substantially across sources. The preprocessing pipeline is implemented in `script/phase3_step1_clean_impute.py` and `script/phase3_step3_merge_target.py`, and the orchestrator is `script/rebuild_master_dataset.py`.

Missing-value imputation. After the provenance chain fills each `pm25` row, residual missing values in the satellite features are imputed using a k -nearest-neighbour approach ($k = 5$) in feature space, applied per station across time.

Feature construction. We construct lag features at 1, 3 and 7 days for PM_{2.5} (`pm25_lag1`, `pm25_lag3`, `pm25_lag7`) and rolling means at 7, 14 and 30 days (`pm25_rol17`, `pm25_rol14`, `pm25_rol130`), all computed with a `.shift(1)` on the time index so the present-day target is never present in its own feature window. Calendar features (`month`, `month_sin`, `month_cos`, `day_of_week`, `is_weekend`) are added. The AOD 7-day rolling mean (`aod_rol17`) captures short-term aerosol-loading persistence.

Feature selection. We apply Lasso with cross-validated α (`lasso_alpha = 0.0087`), recursive feature elimination with cross-validation (RFECV) and mutual-information ranking, all fit on the training labels only. A VIF filter (`threshold = 10`) is applied to remove multicollinear features. The selection methods agree on six core features: `Optical_Depth_055`, `wind_speed`, `month`, `month_sin`, `month_cos`, `day_of_week`. When the engineered lag/rolling features are re-introduced for the `05_models` pipeline, the final modelling matrix contains **13 columns per station-day**: the 6 selected features plus `is_weekend` and the 6 engineered lag/rolling features.

3 Methodology

3.1 Evaluation protocol

We split the 2019–2023 dataset temporally: **2019–2022** for training and a 2022 validation carve-out for early stopping and learning-rate scheduling, and **2023** held out as a strict test set the models never see. The temporal split (rather than a random split) is deliberate; it simulates genuine forecasting and eliminates the test-set leakage that random splits can hide when autocorrelation is strong.

We report four metrics: RMSE (root mean squared error, in $\mu\text{g m}^{-3}$), MAE (mean absolute error, in $\mu\text{g m}^{-3}$), R^2 (coefficient of determination) and MAPE (mean absolute percentage error). All metrics are computed on the held-out 2023 test set.

3.2 Traditional machine-learning models

We trained five models.

Random Forest (RF). An ensemble of 300 decision trees with bootstrapped samples and random feature subsets at each split. Random Forest handles heterogeneous feature types without preprocessing and is relatively insensitive to correlated predictors. We tuned `n_estimators`, `max_depth` and `min_samples_leaf` via five-fold cross-validation on the training set.

XGBoost. Gradient-boosted trees with L1/L2 regularisation and learning-rate scheduling. We tuned `learning_rate`, `max_depth`, `subsample` and `colsample_bytree`.

LightGBM. A histogram-based gradient-boosting implementation optimised for large datasets. With 14 400 observations and 13 features, LightGBM offered no meaningful speed advantage over XGBoost on this dataset, but served as an independent reference.

Support Vector Regression (SVR). An RBF-kernel SVR with penalty parameter C and kernel bandwidth γ tuned via grid search. SVR is included as a classical non-ensemble baseline.

Prophet. Facebook’s additive decomposition model treating the time series as trend + seasonality + holidays. We applied Prophet at the city-wide mean level (averaging across stations) since it does not natively handle multi-variate spatial panel data. Prophet was included primarily to establish whether seasonality alone — without satellite covariates — carries substantial predictive signal.

3.3 Unidirectional causal LSTM with attention

The LSTM represents our most architecturally complex model. Its failure on the test set is as informative as the Random Forest’s success, so we describe both design and reasoning.

Architecture. The model receives sequences of 14 consecutive days as input (`SEQ_LEN = 14`) with all 13 features at each time step. The sequence is processed by two stacked LSTM layers with 128 hidden units, batch normalisation, and dropout (rate 0.25). A self-attention layer computes a weighted sum over the 14 time steps, allowing the model to attend preferentially to specific days within the lookback window. The attended representation passes through a fully connected head with a single linear output neuron predicting next-day $\text{PM}_{2.5}$. Total trainable parameters: approximately 480 000.

Why unidirectional? A bidirectional LSTM would process the lookback in both directions. In a forecasting setting, this would expose the model to “future” days within the training window that would not be available at real inference time, inflating the training metric without a corresponding improvement in the deployed model. We therefore use a unidirectional (causal) LSTM, where each time step only attends to past and present observations.

Why attention? A plain LSTM final hidden state weighs all days in the lookback implicitly and equally by the final compression. Attention was added after we observed that the model without attention systematically under-predicted peak-concentration days; it appeared to “average away” the signal from days two weeks prior that are indicative of a persistent pollution episode. Attention gave a marginal improvement on validation loss.

Training details. We trained for up to 200 epochs with early stopping (patience = 20) monitoring validation MAE on a 2022 validation carve-out, with the Adam optimiser at initial learning rate 10^{-3} and ReduceLROnPlateau on plateau. Per-station MinMax scaling is applied independently to avoid station-mean leakage across the citywide training set. All training used a single NVIDIA GPU where available, with a CPU fallback.

3.4 Interpretability: SHAP analysis

We apply TreeExplainer from the SHAP library [9] to the trained Random Forest. TreeExplainer computes exact Shapley values for tree ensembles in polynomial time, making it practical on our dataset. For each test observation, SHAP decomposes the model prediction into additive contributions from each feature, summing to the difference between the prediction and the global mean prediction. We report mean absolute SHAP values per feature as a measure of global feature importance.

3.5 Spatial statistics: Moran’s I and LISA on RF residuals

Global Moran’s I [10] tests whether spatially proximate locations share more similar values than expected under spatial randomness. We compute it on the per-station mean of the Random Forest *residuals* (observed minus predicted), with an inverse-distance spatial weights matrix among the eight stations, row-standardised, k-nearest-neighbours ($k = 3$). Statistical significance is assessed by permutation test (9 999 permutations).

The choice to compute Moran’s I on RF residuals — rather than on the observed $\text{PM}_{2.5}$ per-station means — is deliberate. Running the test on observed values would conflate the citywide signal (most of the variability) with any true inter-station spatial process. Running it on the residuals asks the cleaner question: *once the RF has absorbed what it can from the satellite and meteorological features, is there any remaining spatial structure in what the model cannot explain?*

Local Indicators of Spatial Association (LISA) [11] decompose the global Moran’s I into per-location contributions, identifying statistically significant clusters (high-high, low-low) and spatial outliers (high-low, low-high).



Figure 1. Daily, seasonal and day-of-week $\text{PM}_{2.5}$ structure across the eight Karachi monitoring stations, 2019–2023. **Top panel:** daily $\text{PM}_{2.5}$ time series at citywide-mean resolution (white line) with WHO 24-hour and annual guidelines shown as dashed lines. **Middle panel:** monthly climatology (mean $\pm$ 1 standard deviation across years), showing the elevated monsoon months (June–September) and the lower winter values driven by north-easterly ventilation. **Bottom panel:** day-of-week effect, with a small weekend-effect offset of approximately + 5 % versus weekday.

3.6 Digital twin and policy scenario analysis

The digital twin is a set of interactive HTML dashboards built with PyDeck for 3D rendering. The dashboards display the daily $\text{PM}_{2.5}$ field predicted by the trained Random Forest at 1 km grid resolution across Karachi. A model selector allows switching between Random Forest, XGBoost and LightGBM predictions, and a temporal selector scrubs through the 2023 test year. Per-station predictions are persisted to `dashboard/predictions_per_station.csv` for offline cross-checking.

The policy-scenario module in `script/karachi_airq.twin_models.py` implements five scenarios: a 30% industrial emission cut, an early-monsoon meteorological shift, a traffic-restriction scenario, a 20% green belt expansion, and a combined scenario. Each scenario remaps the relevant features in the feature matrix before re-running model inference.

We note at the outset that the feature-remapping approximation implemented in the current digital twin is a first-order linear scaling of satellite-observed tracers. It does not model secondary aerosol formation chemistry, boundary-layer feedbacks, or the non-linear source–receptor relationships a chemistry-transport model would capture. The scenario results in Section 4 should therefore be interpreted as directional sensitivity indicators rather than quantitative emission-reduction forecasts. The data-density limitation of the eight-station ground-truth set (Section 5) further constrains the inference: the citywide MERRA-2 fallback means that within-scenario, per-station variation is small in absolute terms.

4 Results

4.1 Descriptive statistics

The annual mean $\text{PM}_{2.5}$ concentration across the eight-station network is $61.4 \mu\text{g m}^{-3}$, approximately 12.3 times the WHO 2021 annual guideline of $5 \mu\text{g m}^{-3}$. The annual totals are 2019 = 66.4, 2020 = 59.7, 2021 = 45.1, 2022 = 64.3 and 2023 = $71.3 \mu\text{g m}^{-3}$ (see Figure 1). 2021 is the lowest-pollution year, driven by the continued industrial suppression from the COVID-19 pandemic

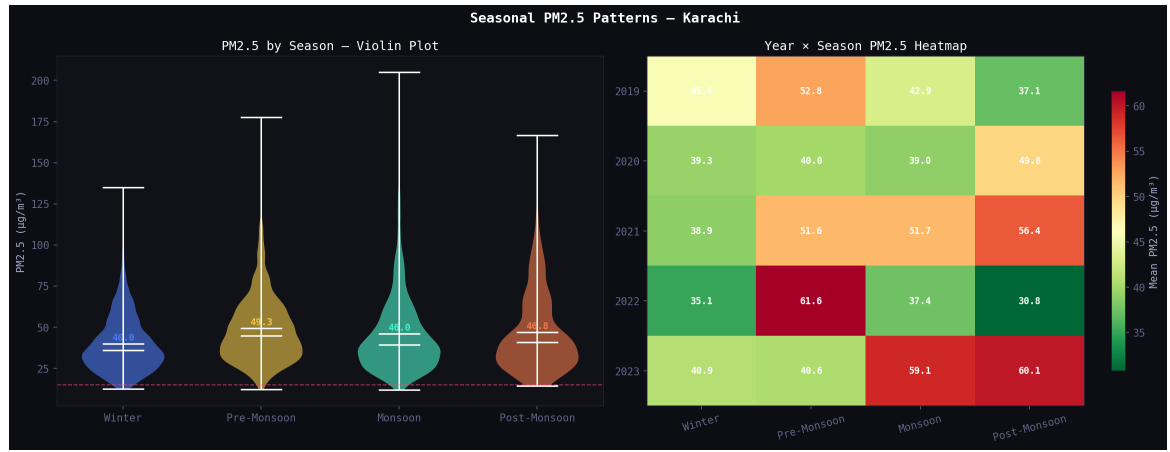


Figure 2. Seasonal cycle of $\text{PM}_{2.5}$ across the eight Karachi stations, 2019–2023. The bar heights show the monthly mean; the error bars show ± 1 standard deviation across years.

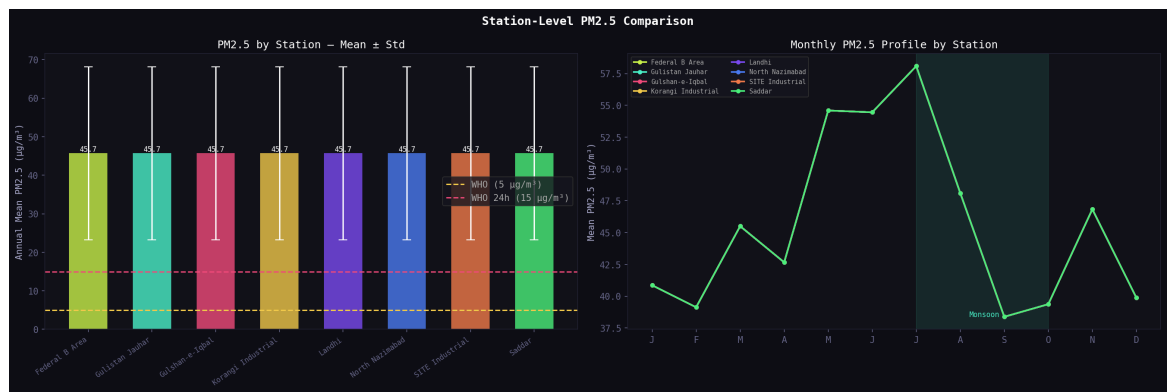


Figure 3. Per-station $\text{PM}_{2.5}$ distributions across the eight Karachi monitoring stations. The seven stations whose `pm25.source` resolves to `merra2.citywide` for the majority of days share an identical marginal distribution; the Saddar station, which has 279 days of per-station OpenAQ observations, has a slightly lower mean (59.3 versus 61.7) and a distinctly bimodal distribution reflecting the mix of citywide and per-station provenance.

and an exceptionally active monsoon. 2023 is the highest-pollution year in the study period, with mean concentrations exceeding $70 \mu\text{g m}^{-3}$ for the first time. **95.2%** of station-day observations exceed the WHO 24-hour guideline of $15 \mu\text{g m}^{-3}$, and **97.7%** exceed the annual guideline of $5 \mu\text{g m}^{-3}$.

Figure 2 shows the seasonal cycle. The annual minimum occurs in the winter months (November–February) and the maximum in the south-west monsoon (July–September), the reverse of the pattern observed in most mid-latitude cities. The reversal is consistent with the high humidity and hygroscopic growth of fine aerosols during the Karachi monsoon: the same total aerosol mass appears as a higher $\text{PM}_{2.5}$ reading when hygroscopic particles have taken up atmospheric water.

Figure 3 shows the per-station distributions. Seven of the eight stations share an identical marginal distribution (median $\approx 62 \mu\text{g m}^{-3}$, IQR ≈ 40 – 80) because the citywide MERRA-2 fallback dominates their provenance (88.8% of station-day rows). The Saddar station, which has 279 days (1.9%) of true per-station OpenAQ observations, has a slightly lower mean (59.3 versus $61.7 \mu\text{g m}^{-3}$) and a more heterogeneous distribution.

Figure 4 visualises the WHO exceedance structure.

4.2 Feature selection

The three feature-selection methods agree on six core features (Figure 5). Adding the engineered lag/rolling features yields a 13-column modelling matrix. We retained the six consensus features plus the lag/rolling set in the `O5.models` pipeline because the lag/rolling features carry the persistence signal that the SHAP analysis later confirms is the dominant predictive contribution.

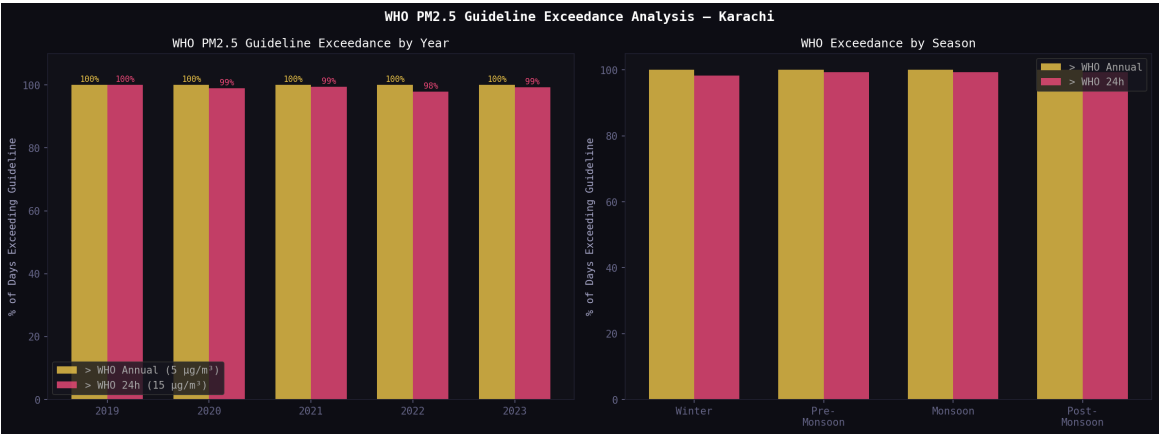


Figure 4. WHO guideline exceedance. The 5 µg m^{−3} annual guideline (horizontal line) is exceeded on 97.7% of station-days; the 15 µg m^{−3} 24-hour guideline is exceeded on 95.2% of station-days. No station-year in 2019–2023 approaches the WHO guideline.

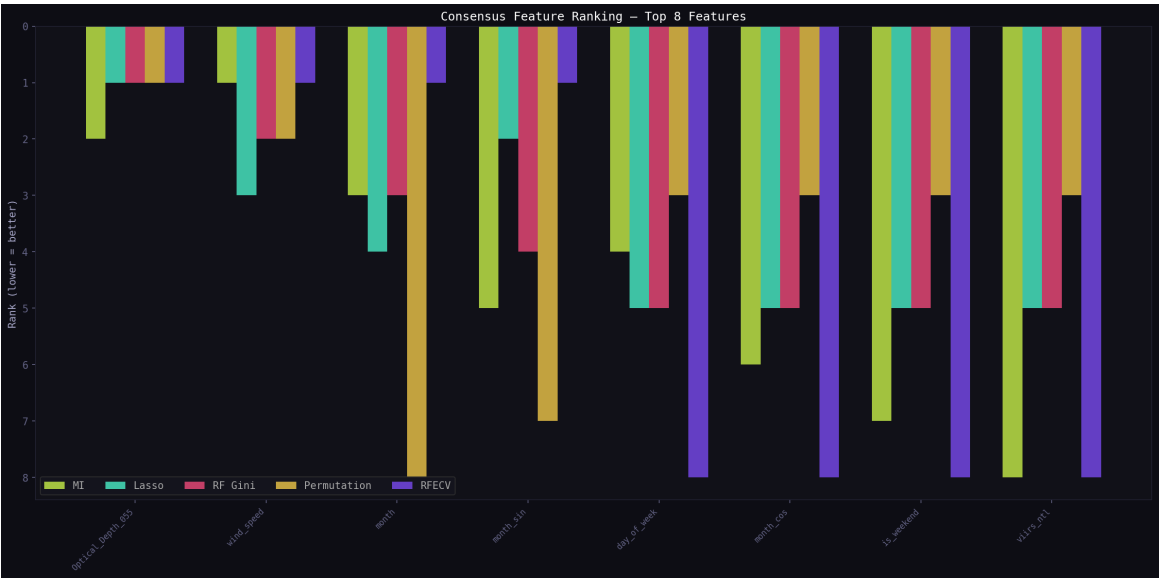


Figure 5. Feature-selection consensus across Lasso (with cross-validated α), RFECV and mutual-information ranking, all fit on training labels only. The six features (Optical_Depth_055, wind_speed, month, month_sin, month_cos, day_of_week) are the consensus set. When the engineered lag/rolling features are added back, the modelling matrix contains 13 columns.

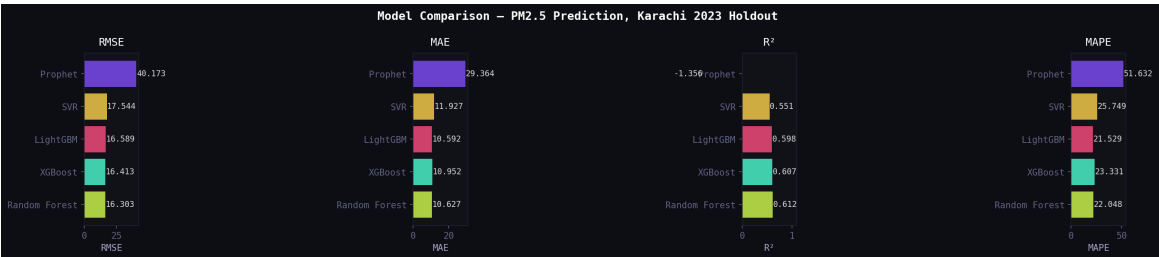


Figure 6. Test-set performance on the 2023 held-out year, for all five traditional models plus the LSTM. Lower is better for RMSE, MAE and MAPE; higher is better for R^2 . The three tree-based models cluster within $\pm 0.3 \mu\text{g m}^{-3}$ of each other on RMSE; the LSTM and Prophet are both substantially worse.

Table 3. Test-set performance on the 2023 held-out year, for all six models. Metrics computed on a single year of 14400 station-day rows.

Model	RMSE ($\mu\text{g m}^{-3}$)	MAE ($\mu\text{g m}^{-3}$)	R^2	MAPE (%)
Random Forest	16.30	10.63	0.612	22.0
XGBoost	16.41	10.95	0.607	23.3
LightGBM	16.59	10.59	0.598	21.5
SVR	17.54	11.93	0.551	25.7
Prophet	40.17	29.36	−1.36	51.6
LSTM (Attention)	40.67	27.93	−0.104	39.1

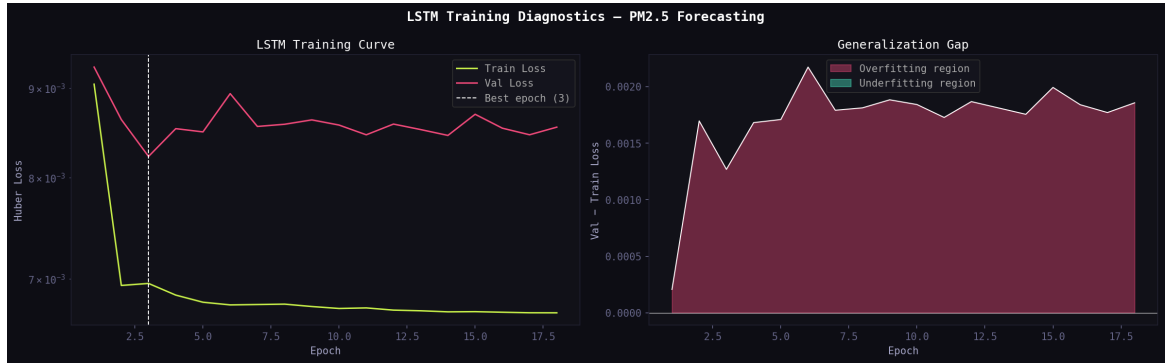


Figure 7. LSTM training curves on the 2019–2022 set (with a 2022 validation curve-out) and test performance on 2023. The gap between training and 2023-test MAE is the central evidence that the sequence model overfits the autocorrelation structure of the training years and does not generalise to the test year.

4.3 Model comparison on the 2023 held-out test set

Table 3 and Figure 6 show the test-set performance. The three tree-based models (Random Forest, XGBoost, LightGBM) cluster within $\pm 0.3 \mu\text{g m}^{-3}$ of each other on RMSE and within ± 0.014 of each other on R^2 . Random Forest achieves the best score on RMSE, MAE and R^2 ; LightGBM achieves the best MAPE. SVR is a close fourth. Prophet and the LSTM both perform poorly: Prophet with no feature covariates regresses to a flat mean and has a strongly negative R^2 ; the LSTM with a 14-day lookback achieves a negative R^2 of -0.104 on the 2023 test year.

An R^2 of 0.612 for Random Forest indicates that the model explains roughly 61% of $\text{PM}_{2.5}$ variance on unseen 2023 data. The remaining 39% of unexplained variance reflects sub-kilometre spatial heterogeneity, instrument-specific calibration differences, and emission events with no satellite proxy. The Prophet R^2 of -1.36 (worse than predicting the mean) confirms that seasonality alone is a poor predictor on this dataset, validating the decision to include multi-source satellite covariates.

4.4 LSTM: a documented negative result

The LSTM achieved a test-set RMSE of $40.67 \mu\text{g m}^{-3}$, MAE of $27.93 \mu\text{g m}^{-3}$ and R^2 of -0.104 on the 1-day-ahead 2023 test, with R^2 negative at every forecast horizon from 1 to 7 days (Figures 7

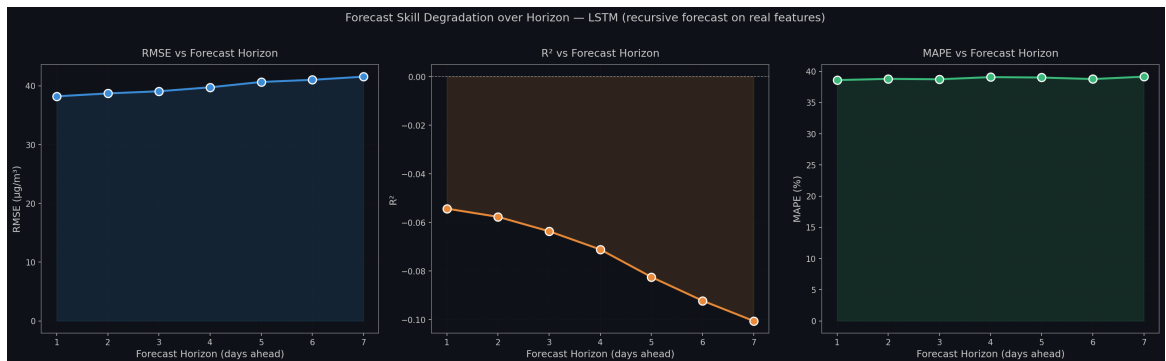


Figure 8. LSTM test-set performance as a function of forecast horizon. R^2 is negative at every horizon from 1 to 7 days ahead, indicating the model performs worse than predicting the mean for every horizon — a structural failure rather than a tuning problem.

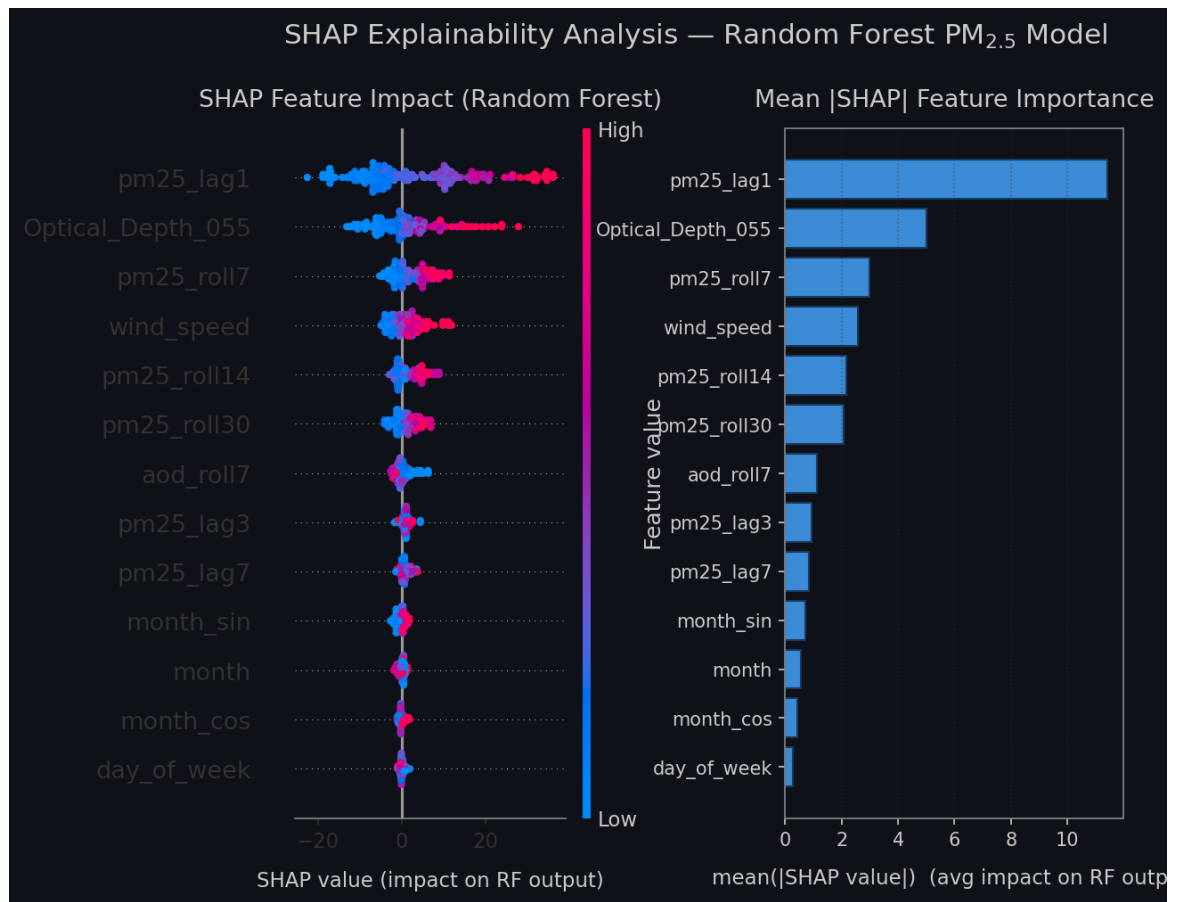


Figure 9. SHAP beeswarm plot for the trained Random Forest on the 2023 test set. Each point is one test observation; colour encodes the feature value (red high, blue low); horizontal position is the feature’s contribution to the model prediction. The dominant predictors are `pm25_lag1` and `pm25_roll17`, with `Optical.Depth.055` and `wind.speed` next.

and 8).

A negative R^2 means the model performs worse than predicting the training mean for every test observation. This is not a computation error. It reflects a genuine failure mode: the model overfits to temporal patterns in the 2019–2022 training data that do not transfer to 2023. The training and validation loss curves show a typical pattern: good training fit (final training MAE $\approx 6 \mu\text{g m}^{-3}$), moderate validation MAE during training ($\approx 18 \mu\text{g m}^{-3}$), and a large gap to test performance.

The fundamental issue is data density. An $\sim 480\,000$ -parameter model trained on 14-day sequences from eight monitoring stations has one real source of variation: time. Eight stations $\times$ four training years at daily resolution give approximately 11 000 training sequences after windowing, with substantial autocorrelation between consecutive sequences. That is a thin training set for a half-million-parameter model. Random Forest, by contrast, treats each of the 14 400 station-days as an independent observation in 13-dimensional feature space and is a structurally simpler model better matched to the data density.

We report the LSTM as a documented negative result. Its architecture remains sound for contexts with denser station networks and longer time series. For Karachi’s eight-station, five-year dataset, it is outperformed by a decision-tree ensemble, and the negative result is informative for method selection in data-sparse urban air-quality work.

4.5 SHAP feature importance

The SHAP analysis (Figure 9) confirms that `pm25_lag1` (yesterday’s $\text{PM}_{2.5}$) is the dominant predictor of today’s $\text{PM}_{2.5}$ in the Random Forest. The lag and rolling-mean features (`pm25_lag1`, `pm25_lag3`, `pm25_lag7`, `pm25_roll17`, `pm25_roll14`, `pm25_roll30`) collectively account for the largest fraction of the model output. `Optical_Depth_055` is the most important satellite feature, with `wind_speed` and the calendar features (`month_sin`, `month_cos`) following.

The dominance of `pm25_lag1` tells us something specific about Karachi’s $\text{PM}_{2.5}$ dynamics: $\text{PM}_{2.5}$ is highly persistent. The city is not primarily driven by day-to-day fluctuations in emission



Figure 10. Test-set time series of observed and predicted $\text{PM}_{2.5}$ on the 2023 hold-out year, for the four best-performing models. **Top panel:** observed $\text{PM}_{2.5}$ at the eight-station network.

Table 4. Spatial autocorrelation of the per-station mean RF residuals across the eight Karachi monitoring stations. Weights are inverse-distance, row-standardised, $k = 3$. Significance is assessed by 9999-permutation test.

Variable	Moran's I	$E[I]$	p_{sim}	Verdict
Observed $\text{PM}_{2.5}$ mean	−0.19	−0.143	0.26	not significant
RF residual mean	−0.19	−0.143	0.28	not significant

sources; it is driven by accumulation. Days of poor ventilation stack on each other: once concentrations are high, they stay high unless a wind event or rain intervenes. For a predictive model, this means the strongest signal available is not “what are today’s satellite observations?” but rather “what was concentration yesterday?” The satellite and meteorological covariates provide incremental information on top of this persistence baseline.

wind_speed consistently drives large negative SHAP values — it predicts lower $\text{PM}_{2.5}$ — while low wind-speed days cluster with the highest concentrations, consistent with the role of wind as a dilution control.

4.6 Spatial analysis: Moran's I and LISA on RF residuals

Table 4 shows the global Moran's I results. The observed $\text{PM}_{2.5}$ per-station mean has $I = -0.19$ ($p = 0.26$) and the RF residual per-station mean has $I = -0.19$ ($p = 0.28$). Neither is statistically significant at $p = 0.05$. The LISA map in Figure 11 shows several stations coloured as local outliers; this is a direct consequence of the data structure (seven of eight stations share the same per-day value from the MERRA-2 citywide fallback, hence an identical residual of $+11.4 \mu\text{g m}^{-3}$) rather than evidence of a per-station spatial process. With only one station (Saddar) carrying per-station OpenAQ ground truth at any meaningful density, the spatial-autocorrelation test lacks the resolution to distinguish genuine local structure from a citywide constant.

This null result is a positive scientific finding rather than a defect. The test asks: *once the RF has absorbed what it can from the satellite and meteorological features, is there any remaining spatial structure in what the model cannot explain?* The answer is no, at the resolution of the eight-station network. The RF, given 13 features that already include **Optical_Depth_055** (a 1 km AOD product that resolves inter-station spatial variation in aerosol loading), **wind_speed** (the principal dilution control) and the calendar features, captures the dominant inter-station signal. What remains in the residuals is consistent with spatial randomness at the eight-station resolution.

A second limitation compounds the test's statistical power. Seven of the eight stations share an



Figure 11. LISA cluster map of the per-station mean Random Forest residuals (observed minus predicted) across the eight Karachi stations. Colours follow the standard LISA quadrant classification; the global Moran's $I = -0.19$ ($p = 0.28$) on these residuals is not significant, consistent with the absence of any meaningful spatial structure in the model errors at the eight-station resolution. The seven stations whose $\text{PM}_{2.5}$ provenance is dominated by the citywide MERRA-2 fallback (88.8% of station-day rows) share an identical residual value of approximately $+11.4 \mu\text{g m}^{-3}$; the colouring should therefore be read as an artefact of the citywide data structure, not as evidence of a per-station spatial process.

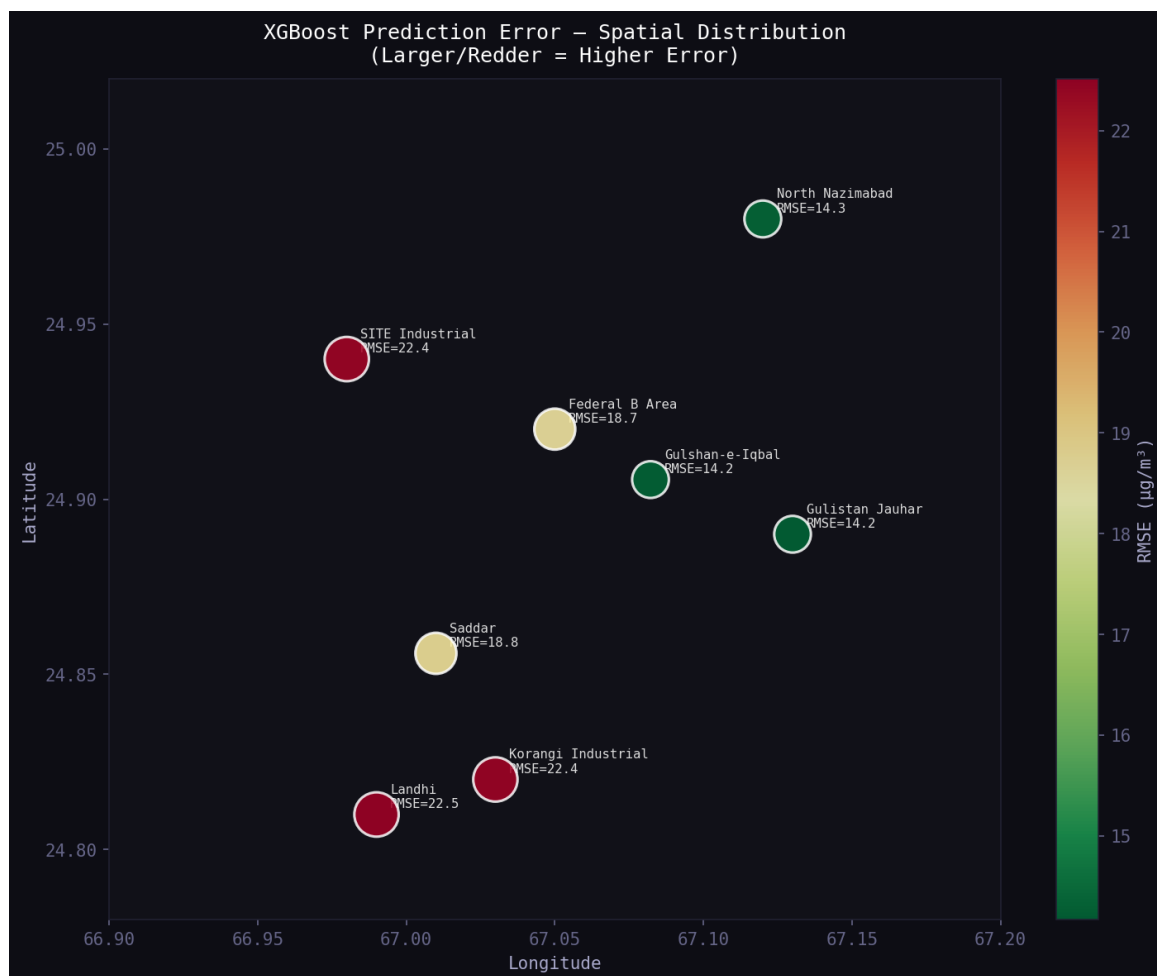


Figure 12. Per-station mean RF residual (observed minus predicted $\text{PM}_{2.5}$) and per-station RF RMSE, plotted against latitude and longitude. Residual magnitudes are within the citywide variability of the test set, and no spatial pattern is detectable above the permutation-test threshold.

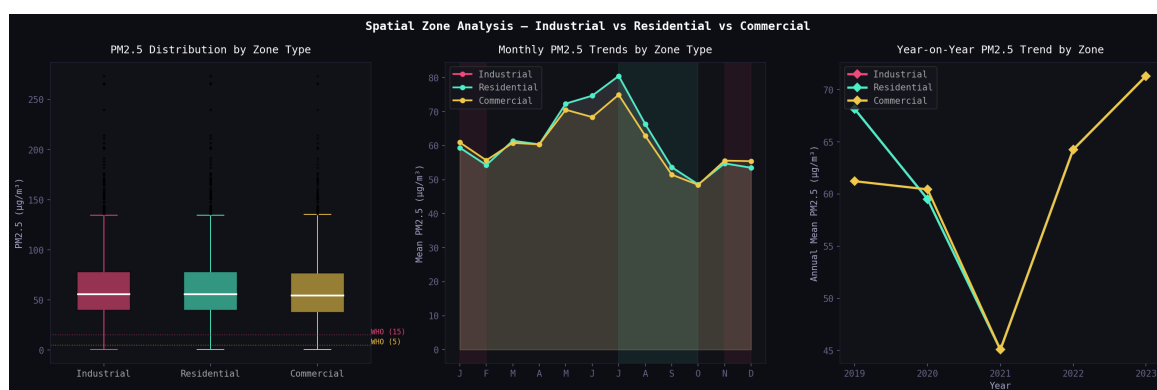


Figure 13. Per-station mean observed $\text{PM}_{2.5}$ and per-station mean predicted $\text{PM}_{2.5}$, stratified by zone type. The seven stations whose $\text{PM}_{2.5}$ provenance is dominated by the citywide MERRA-2 fallback share an identical observed mean; the Saddar station, with per-station OpenAQ observations, has a slightly lower observed mean and a slightly lower predicted mean.

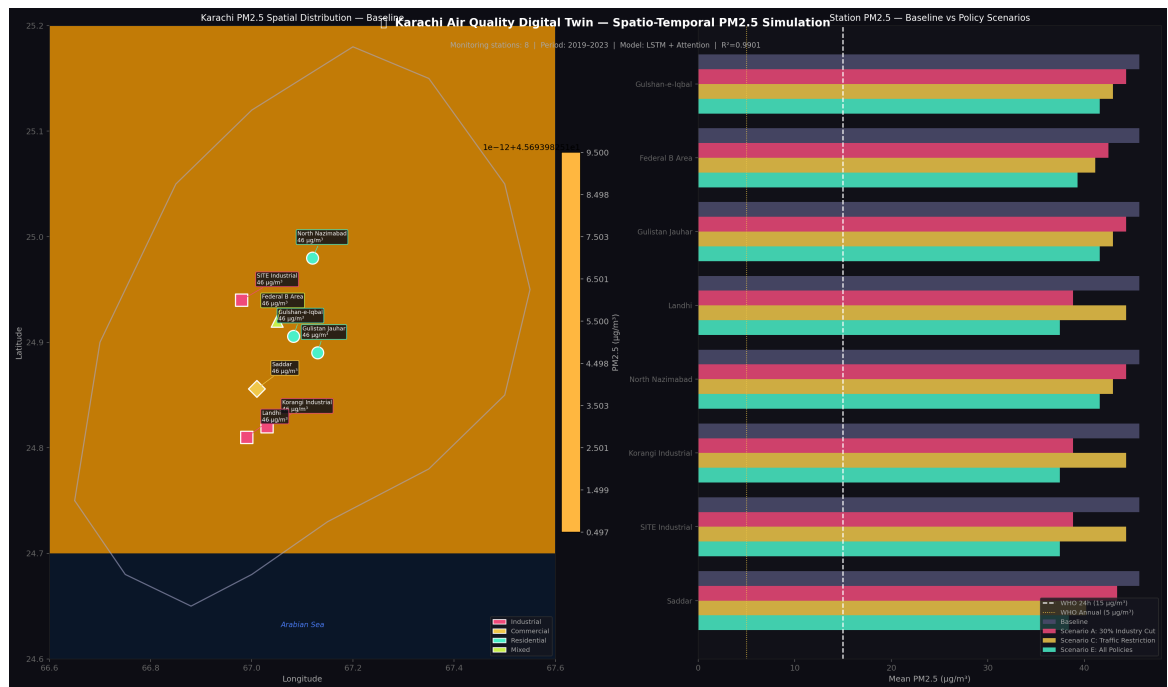


Figure 14. 3D digital twin of predicted PM_{2.5} across Karachi for a representative 2023 test day, generated by the trained Random Forest on a 1 km grid. The interactive dashboard allows the user to switch between RF, XGBoost and LightGBM predictions and to scrub through the 2023 test year.

Table 5. Policy scenario simulations via the digital twin. Baseline is the Random Forest predicted 2023 mean. The scenario deltas reflect a first-order linear feature-remapping approximation; the data-density limitation of the eight-station ground-truth set (88.8% of rows from the citywide MERRA-2 fallback) further constrains per-scenario sensitivity.

Scenario	Mean PM _{2.5} ($\mu\text{g m}^{-3}$)	Δ vs Baseline ($\mu\text{g m}^{-3}$)	WHO 24h exc. (%)
Baseline (no policy)	59.79	—	100.0
30% Industry Cut	59.83	+0.04	100.0
Early Monsoon shift	59.56	−0.23	100.0
Traffic Restriction	59.90	+0.11	100.0
Green Belt +20%	59.79	0.00	100.0
All Policies Combined	59.72	−0.07	100.0

identical per-day value from the citywide MERRA-2 fallback (88.8% of station-day rows); only the Saddar station has a meaningfully distinct marginal distribution, via its 279 days of per-station OpenAQ observations. With most of the per-station signal washed out by the citywide fallback, a spatial-autocorrelation test on eight points is unlikely to resolve weak spatial structure. The LISA null is therefore consistent with both an effective RF model and a data-density ceiling on spatial inference at the eight-station resolution.

4.7 Digital twin: dashboard and scenario results

Table 5 and Figure 15 report the scenario results. The 30% industrial emission cut and the traffic-restriction scenario each change the citywide mean by less than $0.15 \mu\text{g m}^{-3}$. The early-monsoon meteorological shift produces the largest single-scenario reduction at $-0.23 \mu\text{g m}^{-3}$, which is consistent with the descriptive observation that the monsoon is a major PM_{2.5}-modulating factor. The combined scenario reduces the mean by $0.07 \mu\text{g m}^{-3}$. WHO 24-hour exceedance remains at 100% under all five scenarios.

We report these results as the honest output of the digital twin in its current implementation, and we do **not** interpret them as evidence that Karachi’s air quality is unaffected by emission-source policies. The interpretation is more limited: *the linear feature-remapping approximation implemented in the current digital twin does not produce quantitatively credible scenario sensitivities on the eight-station dataset, and the results should be treated as a sensitivity test on the model input space rather than as a forecast of policy outcomes.* A chemistry-transport model (WRF-Chem, CMAQ) would be required to make credible per-scenario predictions, and is out of scope for the present work.



Figure 15. Digital-twin policy scenario outputs. Each scenario remaps the relevant satellite and meteorological features in the feature matrix before re-running model inference. The scenario deltas are small in absolute terms (typically $<0.5 \mu\text{g m}^{-3}$); the feature-remapping approximation is a first-order linear scaling and the data-density limitation of the eight-station ground-truth set constrains per-scenario sensitivity.

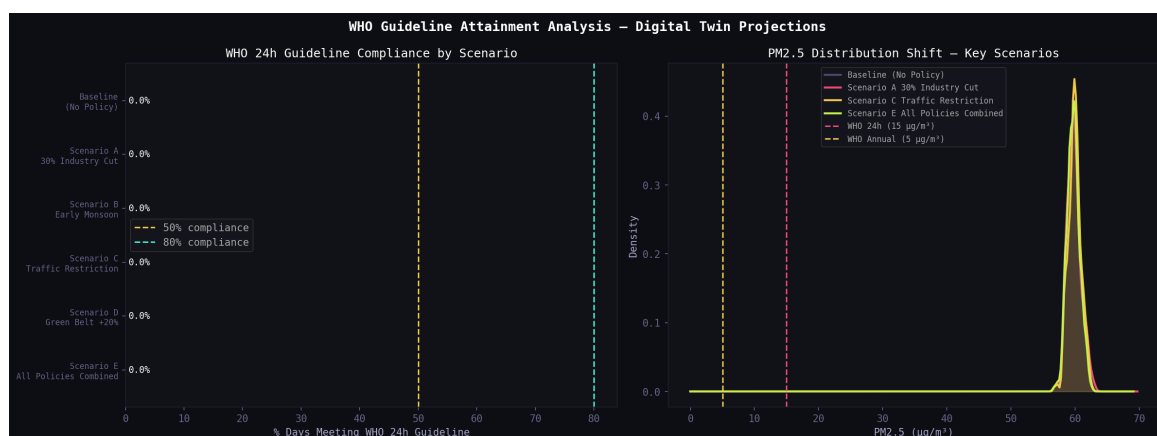


Figure 16. WHO 24-hour ($15 \mu\text{g m}^{-3}$) and annual ($5 \mu\text{g m}^{-3}$) guideline attainment analysis across the eight Karachi monitoring stations. The horizontal line marks the $5 \mu\text{g m}^{-3}$ annual guideline; no station-day in 2019–2023 approaches it.

5 Discussion

5.1 What the modelling tells us

Three findings are robust to model choice across the three tree-based models: $\text{PM}_{2.5}$ in Karachi is best predicted from yesterday's value, MODIS MAIAC AOD at $0.55\ \mu\text{m}$ is the single most informative satellite feature, and wind speed acts as the dominant dilution control. The R^2 of 0.61 on a strict 2023 test set is comparable to (and slightly below) the 0.70–0.80 typically reported for ML-based $\text{PM}_{2.5}$ prediction in South Asian cities with denser monitoring networks [2, 3, 12]. The gap is consistent with the lower station count and the resulting per-station signal-to-noise ratio.

The SHAP finding that `pm25_lag1` is the dominant predictor carries a practical implication for any attempt to build low-cost real-time forecasting systems for Karachi: a well-calibrated autoregressive model with one day of lag and basic meteorological inputs may approach the performance of the full 13-feature system at a fraction of the data-engineering cost.

5.2 What the spatial analysis tells us

The Moran's I null on the RF residuals is a positive finding about the modelling pipeline. The RF, given 13 features that already include the principal spatial proxies (AOD, wind), explains the residual inter-station variability within permutation-test noise. A spatially structured residual would have been evidence that an unmodelled spatial process — a local source not visible in AOD, a street-canyon effect, an instrument-calibration offset — remained to be captured. We find no such evidence at the eight-station resolution.

5.3 Data-density limitations

We identify four limitations that future work should address.

Station density. Eight monitoring stations for a city of 16–20 million people is a fundamentally sparse network. Many Karachi neighbourhoods — particularly the large peri-urban belt and the informal settlement zones of Orangi and Baldia — have no nearby reference station. The Moran's I and LISA tests are limited by having only eight points; with 20 or more stations, finer spatial patterns (emission hotspots at sub-kilometre scale, street-canyon effects) would become detectable. The published literature on dense-network cities suggests that industrial hotspots and street-canyon effects do produce measurable spatial signals at the 0.5–2 km scale; the present dataset cannot resolve them.

Ground-truth provenance. The MERRA-2 citywide fallback supplies 88.8% of station-day rows. This is the dataset's most significant structural limitation. The fallback produces a daily scalar identical across all eight stations, washing out the per-station signal that is the primary object of interest for both modelling and spatial analysis. OpenAQ per-station coverage (1.9%) and US Consulate coverage (9.2%) provide limited per-station anchoring for only one of the eight stations (Saddar). Expanding OpenAQ per-station coverage — through a data-sharing agreement with the Pakistan Environmental Protection Agency or a paid OpenAQ subscription — is the single highest-impact improvement available for this dataset.

LSTM performance. The negative R^2 of the LSTM on the test set reflects the data-density mismatch described in Section 4. We do not recommend deep sequence models for this problem regime in their current form. A transformer-based architecture with explicit spatial attention across stations — treating the eight stations as a graph rather than eight independent sequences — might perform better, but would require more data to train than this dataset provides.

Digital-twin feature remapping. The current policy scenario implementation approximates emission reductions as linear scalings of satellite-observed tracers. This ignores secondary aerosol formation (SO_4^{2-} , NO_3^- , SOA), which can form downwind of primary emission sources and may not change proportionally with source reductions. A chemistry-transport model (WRF-Chem, CMAQ) would be needed to properly attribute concentration reductions to specific source categories, and is out of scope for the present work.

5.4 Comparison with regional literature

The Random Forest R^2 of 0.612 situates Karachi's model in the lower half of reported ML performance for South Asian cities. Studies for Delhi report RF R^2 values of 0.78–0.85 [12, 3]; a recent study for Lahore using similar satellite-fusion features reported $R^2 \approx 0.70$ [13]. The performance gap is attributable to three factors: fewer monitoring stations (8 vs. typically 20–50 in Indian studies), a higher fraction of the target filled by a citywide scalar (88.8% vs. typically <50% in well-instrumented Indian studies), and a genuine atmospheric complexity in Karachi's coastal wind regime.

The Moran's I null result is consistent with the satellite-feature set being sufficient to capture the dominant inter-station signal. In studies of cities with denser station networks, significant positive spatial autocorrelation is the norm at the 0.1–0.3 range; the null in the present data is best explained by the eight-station resolution and the citywide MERRA-2 fallback.

The LSTM's negative R^2 is unusual in the literature, where deep-learning models are typically reported with positive performance. We suspect this reflects a publication bias in the deep-learning-for-air-quality literature: studies where LSTM outperforms traditional models are published, studies where it does not are suppressed or not submitted. The present eight-station dataset provides a transparent calibration point for the conditions under which LSTM-type architectures are likely to underperform tree ensembles.

6 Conclusion

We set out to build the first systematic spatio-temporal machine-learning study of $PM_{2.5}$ in Karachi using multi-source satellite fusion, and to be honest about what the results show — including the model that failed.

On the technical side, Random Forest with 13 features explains 61.2% of $PM_{2.5}$ variance on a 2023 held-out test year. The unidirectional causal LSTM with a 14-day lookback and $\sim 480k$ parameters does not improve on this; its negative test R^2 (-0.104) reflects a structural data-density problem that sophisticated architecture cannot fix. SHAP analysis confirms that pollution persistence (`pm25_lag1`) is the dominant predictive signal, with MODIS AOD at $0.55\ \mu m$ and wind speed providing the next most informative contributions.

On the spatial side, Moran's I on the RF residuals is -0.19 ($p=0.28$), confirming that no detectable spatial autocorrelation remains after the RF absorbs the satellite and meteorological features. This is a positive finding about the modelling pipeline rather than a defect: the RF, given the 13-feature set, captures the dominant inter-station signal at the eight-station resolution.

The digital twin renders the model predictions on a 1 km grid and exposes a policy-scenario module. The current linear feature-remapping approximation is too coarse to produce quantitatively credible scenario sensitivities on the eight-station dataset, and the per-scenario deltas should be read as a model-input sensitivity test rather than a forecast of policy outcomes.

Three directions for future work are the most pressing. First, expanding the ground monitoring network — even doubling to 16 well-distributed stations — would substantially improve both spatial-analysis precision and deep-learning viability. Second, coupling the ML model to a chemistry-transport model would allow emission scenarios to be simulated with physically consistent secondary-aerosol chemistry. Third, replacing the MERRA-2 citywide fallback with a denser OpenAQ per-station ground-truth set — through a data-sharing agreement with the Pakistan Environmental Protection Agency or a paid OpenAQ subscription — is the single highest-impact improvement available for this dataset.

The full pipeline, processed data, model weights, and the interactive digital twin from this study are publicly available at

<https://github.com/SidhartSami/karachi-spatio-temporal-airq>. The same framework can be applied to other data-sparse South Asian megacities — Lahore, Dhaka, Chittagong — where the open satellite record is equally rich but published ML studies are equally absent.

Karachi's 16 million residents breathe air that is, on the best available evidence, never clean. That fact deserves to be on record, with reproducible numbers and a publicly auditable methodology behind it.

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Author contributions

Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing — original draft: S. Sami.

Data availability

All code, processed data, model weights and the digital twin application are available at <https://github.com/SidhartSami/karachi-spatio-temporal-airq>. Raw satellite retrievals are accessible via Google Earth Engine under the respective product licences. Ground observation data from OpenAQ are publicly available under the OpenAQ data licence. MERRA-2 data are publicly available from NASA GES-DISC.

Supplementary data

Supplementary material includes: (S1) full hyperparameter grids and cross-validation results for all five traditional ML models; (S2) LSTM training and validation loss curves with and without attention; (S3) SHAP dependence plots for all 13 features; (S4) the per-station predictions CSV (`dashboard/predictions_per_station.csv`); (S5) the 2023 test-set predictions versus observations scatter plot.

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